Exp No: 2	Study of Network cables - straight-through and crossover ca

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AIM:

To study the different types of network cables and to make a straight-through cable and a cross-wired (crossover) cable using a crimping tool.

APPARATUS REQUIRED:

- Cat5e / Cat6 UTP cable (about 1 metre for each cable)
- RJ45 connectors (4 numbers)
- Crimping tool with built-in cutter and stripper
- LAN cable tester (main unit and remote unit)

ALGORITHM / PROCEDURE:

Part A - Study of network cables

1. Observe the sample twisted pair, coaxial and fibre optic cables available in the lab.
2. Note the outer jacket marking (for example Cat5e, 24 AWG) and the number of pairs.
3. Write down the use, speed and maximum length of each cable type in the table given below.

Part B - Making the cables

1. Cut the UTP cable to the required length (about 1 m) using the cutter.
2. Remove about 2.5 cm of the outer jacket from both ends with the stripper. Take care not to cut the inner wires.
3. Untwist the four pairs and straighten the eight wires with fingers.
4. For the **straight-through cable**: arrange the wires in `T568B` order on both ends.
5. For the **crossover cable**: arrange `T568B` order on one end and `T568A` order on the other end.
6. Cut the wire tips even so that about 1.3 cm of the wires is left outside the jacket.
7. Hold the RJ45 connector with the clip facing down, push the wires inside till they touch the end of the connector. The jacket should also go inside a little.
8. Put the connector in the crimping tool and press firmly so that the pins cut into the wires.
9. Repeat steps 2 to 8 for the other end of the cable.
10. Insert both ends into the LAN tester and check the LED sequence.

Types of network cables

Cable	Connector	Max. length	Speed	Used for
Twisted pair (UTP / STP)	RJ45	100 m	10 Mbps to 10 Gbps	LAN, connecting PCs to switch
Coaxial (RG-6, RG-59)	BNC / F-type	185 m to 500 m	10 Mbps (old Ethernet)	Cable TV, CCTV, old LANs
Fibre optic (single / multi mode)	SC / LC / ST	2 km (multi) up to 40 km+ (single)	1 Gbps to 100 Gbps	Backbone, long distance links

UTP category	Max. speed	Bandwidth
Cat5	100 Mbps	100 MHz

UTP category	Max. speed	Bandwidth
Cat5e	1 Gbps	100 MHz
Cat6	1 Gbps (10 Gbps up to 55 m)	250 MHz
Cat6a	10 Gbps	500 MHz

Pin	1	2	3	4	5	6	7	8
T568A	W/G	G	W/O	B	W/B	O	W/Br	Br
T568B	W/O	O	W/G	B	W/B	G	W/Br	Br

W = white stripe, G = green, O = orange, B = blue, Br = brown

Cable	Wiring	Used to connect
Straight-through	T568B to T568B	PC to Switch, PC to Hub, Router to Switch (unlike devices)
Crossover	T568B to T568A	PC to PC, Switch to Switch, Hub to Hub, PC to Router (like devices)

DIAGRAM:

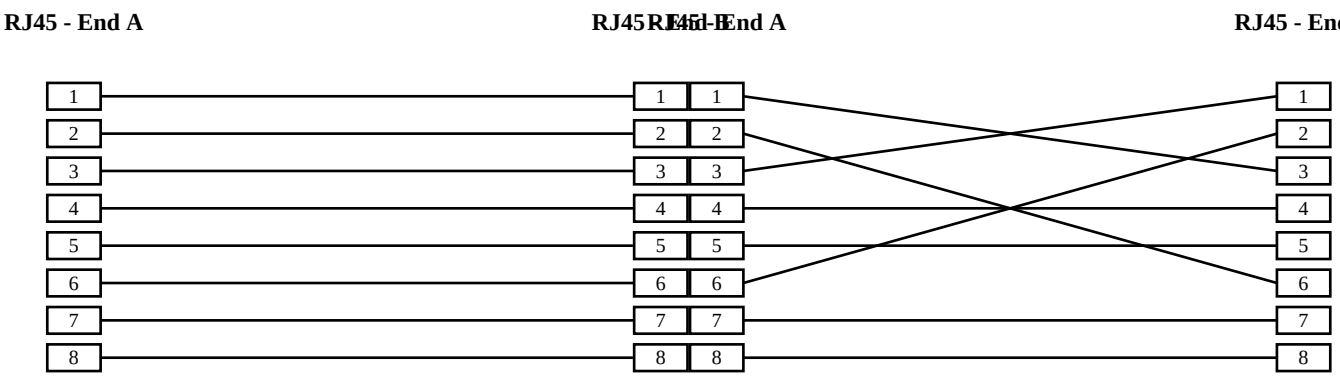
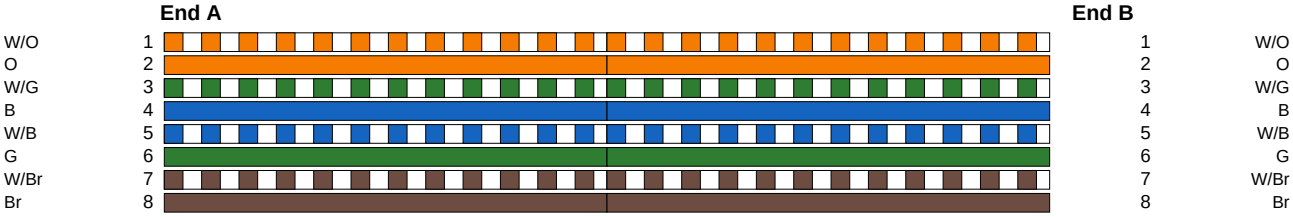


Fig 1: Straight-through cable (pin 1 to 1, 2 to 2 ...)

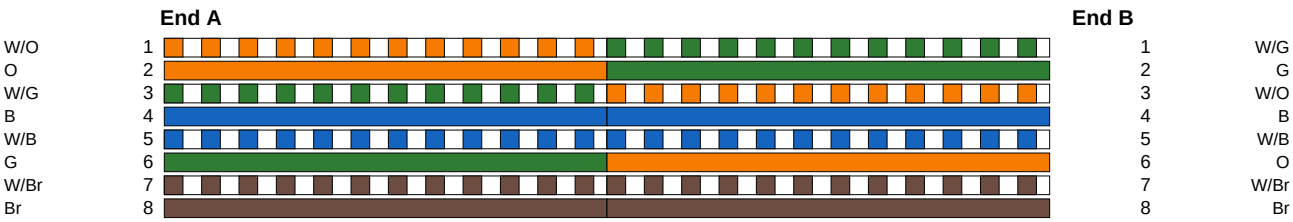
Fig 2: Crossover cable (1 to 3, 2 to 6, 3 to 1, 6 to 2)

OUTPUT:

Colour order of the wires at both ends

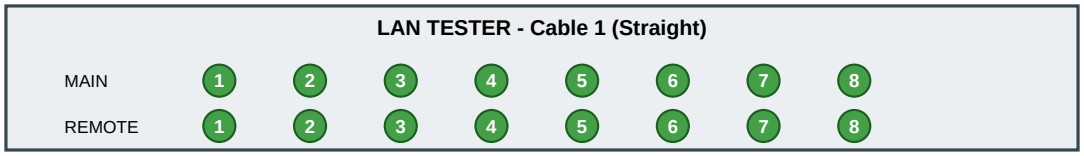


Cable 1: Straight-through cable (T568B - T568B)

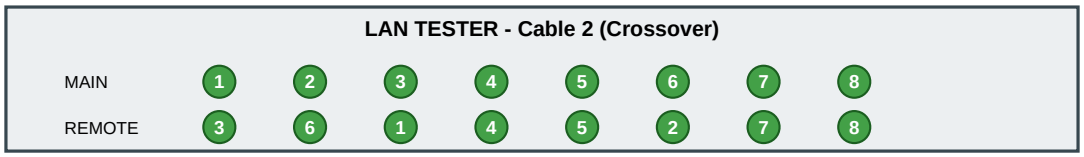


Cable 2: Crossover cable (T568B - T568A)

LAN cable tester result



LEDs glow in the order 1-2-3-4-5-6-7-8 : cable is OK



LEDs glow in the order 3-6-1-4-5-2-7-8 : crossover cable is OK

Cable	Main unit LED	Remote unit LED	Status
Straight-through	1 2 3 4 5 6 7 8	1 2 3 4 5 6 7 8	Working
Crossover	1 2 3 4 5 6 7 8	3 6 1 4 5 2 7 8	Working

RESULT:

Thus the different types of network cables were studied. A straight-through cable and a crossover cable were made using the crimping tool and both were tested successfully with the LAN tester.